

AAA

Review Video: <https://coreultrasound.com/aaa/>

Background – POCUS for detecting AAA

- Sensitivity 99%
- Specificity 99%

Knobology/Preparation:

- Supine with abdomen exposed
- Depth: **30cm** (penetration)
- Gain: mid-range
- Scan required from the **xiphoid to the iliac bifurcation**.
- Perform in the **transverse plane**
- Probe: **curved array**
- Systems preset: aorta /abdominal

External Landmark: **xiphoid process**

Internal Landmark: **spine**

AOI: **echogenic wall of the aorta**

STEPS:

1. Place the probe just **caudal to the xiphoid process (external landmark)** in **transverse plane**
 - Make sure to be totally transverse, otherwise there will be excess attenuation, and it will be very hard to see the aorta.
2. Identify the **spine (internal landmark)**, this is the landmark for the aorta
 - Hyperechoic structure with acoustic shadowing past it
3. Identify the **aortic wall**, this is the **area of interest**
 - Aorta:
 - **Location:** *immediately anterior to the spine, usually on the (patient) left in comparison to the IVC*
 - **Wall thickness:** *aorta wall is thicker than that of the IVC - significantly more echogenic.*
 - **Non-compressibility:** *if you push into the belly with the probe, the aorta should NOT collapse, whereas the IVC will.*
 - **Lack of respiratory variability:** *IVC will collapse when the patient inspires (negative chest pressure draws blood into the chest and momentarily depletes it).*
 - **Doppler**
4. Once identified, center and then magnify the aorta.
 - Continue to optimize depth as you continue with your scan.
5. Once image is optimized, slide the probe down the abdomen caudally until you see the bifurcation of the aorta
 - It is critical to keep the probe at a 90-degree angle to the skin, otherwise you can falsely increase/decrease the lumen size and over/under call AAA.

IVC	Aorta
•Patient's RIGHT •Enters abdomen HIGH •Compressible, ovoid •Thinner walls •No anterior branches below the hepatic veins •Larger (may depend on hydration status) •Usually respiratory variation •Pulsatile (both respiratory AND cardiac)	•Patient's LEFT •Enters abdomen LOW •Non-compressible, round, and thick walls •Anterior branches •Smaller (unless AAA) •No respiratory variation •Pulsatile

*Normal aortic calibre is **3cm** from **outer wall to outer wall**- can use the calipers to make formal measurements should they be required.*

Troubleshooting

- Flex hips/knees
- Use firm probe pressure and hold
- Ask the patient to take a deep breath in and hold it / exhale it – should help move bowel gas (works upper 1/3 to 1/2 of abdomen)
- Slide laterally and heel medially to move around loops of bowel
- If significant adipose tissue, consider lowering probe frequency

Pitfalls

- **Image interpretation**
 - **Chronic Thrombus:** measuring the diameter of the false lumen, instead of the true lumen which is outer wall to outer wall - you have to measure all of it, including the more echogenic thrombus
 - Saccular aneurysms can be easily overlooked, so make sure to be methodical with your scan.
- **Image generation**
 - Scanning too fast
 - Not scanning in true transverse
 - Poor probe control/grip
 - Not starting at the xiphoid process
 - Overestimating size by taking oblique cut
 - Not visualizing 1cm or more of the aorta due to bowel gas
- **Clinical integration**
 - Assuming a AAA < 5cm cannot rupture
 - Assuming the debris in AAA represents thrombus and therefore means acute rupture.
 - Declaring negative scan without having visualized the entire aorta

Determinant Positive Scan: if you see the diameter enlarge past 3cm, don't stop - find the largest area of the diameter. After scanning the whole thing, go back to the largest diameter and measure with calipers Most AAAs are fusiform in shape - pick the diameter that looks biggest in size - **measure from outer wall to outer wall.**

Determinant Negative Scan: evaluation of the outer wall of the abdominal aorta from the xiphoid process to the iliac bifurcation where at no point the outer wall exceeds 3cm.

Indeterminate scans: When you cannot visualize the entire abdominal aorta from the xiphoid process down to the bifurcation - gas, obesity can very often get in the way and cause significant scatters